JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON FRIDAY 23rd JANUARY 2026) **TIME : 3:00 PM TO 6:00 PM**

MATHEMATICS — TEST PAPER WITH ANSWER

SECTION-A

1. If $f(x)=\begin{cases}\dfrac{a|x|+x^2-2(\sin|x|)(\cos|x|)}{x} & , x\neq 0\\ b & , x=0\end{cases}$

is continuous at x = 0, then a + b is equal to :

(1) 1 (2) 2

(3) 0 (4) 4

2. Let $\frac{\pi}{2}<\theta<\pi$ and $\cot\theta=-\frac{1}{2\sqrt{2}}$. Then the value of

$$\sin\left(\frac{15\theta}{2}\right)(\cos 8\theta+\sin 8\theta)+\cos\left(\frac{15\theta}{2}\right)(\cos 8-\sin 8\theta)$$

is equal to :

(1) $\frac{1-\sqrt{2}}{\sqrt{3}}$ (2) $-\frac{\sqrt{2}}{\sqrt{3}}$

(3) $\frac{\sqrt{2}-1}{\sqrt{3}}$ (4) $\frac{\sqrt{2}}{\sqrt{3}}$

3. Let $\vec{a}=\hat{i}-2\hat{j}+3\hat{k}$, $\vec{b}=2\hat{i}+\hat{j}-\hat{k}$, $\vec{c}=\lambda\hat{i}+\hat{j}+\hat{k}$ and $\vec{v}=\vec{a}\times\vec{b}$. If $\vec{v}.\vec{c}=11$ and the length of the projection of $\vec{b}$ on $\vec{c}$ is p, then $9p^2$ is equal to :

(1) 9 (2) 6

(3) 4 (4) 12

4. Let A(1, 2) and C(–3, –6) be two diagonally opposite vertices of a rhombus, whose sides AD an BC are parallel to the line 7x – y = 14. If B (α, β) and D(γ, δ) are the other two vertices, then $|\alpha+\beta+\gamma+\delta|$ is equal to :

(1) 9 (2) 3

(3) 6 (4) 1

5. Consider two sets A = {x∈z : |(|x – 3| – 3)|≤1} and

$$B=\left\{x\in\mathbb{R}-\{1,2\}:\frac{(x-2)(x-4)}{x-1}\log_e(|x-2|)=0\right\}.$$

Then the number of onto functions f : A→B is equal to :

(1) 62 (2) 79

(3) 32 (4) 81

6. The system of linear equations

x + y + z = 6

2x + 5y + az = 36

x + 2y + 3z = b

has

(1) unique solution for a = 8 and b = 16

(2) infinitely many solutions for a = 8 and b = 14

(3) infinitely many solutions for a = 8 and b = 16

(4) unique solution for a = 8 and b = 14

7. The sum of all the real solutions of the equation $\log_{(x+3)}(6x^2+28x+30)=5-2\log_{(6x+10)}(x^2+6x+9)$ is equal to :

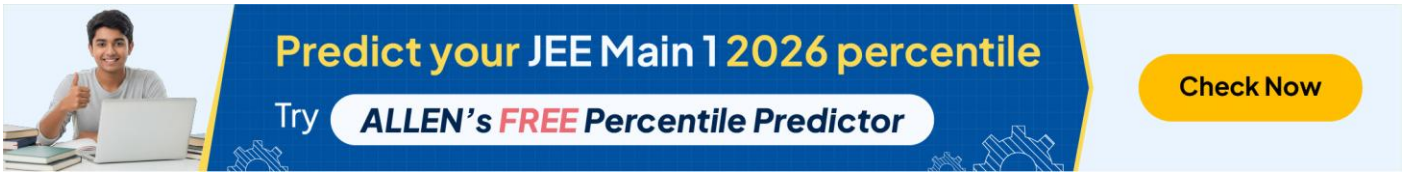
(1) 2 (2) 1

(3) 0 (4) 4

8. If the points of intersection of the ellipses $x^2+2y^2-6x-12y+23=0$ and $4x+2y^2-20x-12y+35=0$ lie on a circle of radius r and centre (a, b), then the value of $ab+18r^2$ is

(1) 53 (2) 51

(3) 52 (4) 55



9. Let PQ be a chord of the hyperbola $\frac{x^2}{4}-\frac{y^2}{b^2}=1$, perpendicular to the x-axis such that OPQ is an equilateral triangle, O being the centre of the hyperbola. If the eccentricity of the hyperbola is $\sqrt{3}$, then the area of the triangle OPQ is :

(1) $2\sqrt{3}$ (2) $\frac{8\sqrt{3}}{5}$

(3) $\frac{11}{5}$ (4) $\frac{9}{5}$

10. Let $\vec{a},\vec{b},\vec{c}$ be three vectors such that $\vec{a}\times\vec{b}=2(\vec{a}\times\vec{c})$. If $|\vec{a}|=1,|\vec{b}|=4$, $|\vec{c}|=2$, and the angle between $\vec{b}$ and $\vec{c}$ is 60°, then $|\vec{a}.\vec{c}|$ is :

(1) 2 (2) 4

(3) 0 (4) 1

11. Bag A contains 9 white and 8 black balls, while bag B contains 6 white and 4 black balls. One ball is randomly picked up from the bag B and mixed up with the balls in the bag A. Then a ball is randomly drawn from the bag A. If the probability, that the ball drawn is white, is $\frac{p}{q}$, gcd(p, q) = 1, then p + q is equal to

(1) 22 (2) 23

(3) 24 (4) 21

12. The area of the region enclosed between the circles $x^2+y^2=4$ and $x^2+(y-2)^2=4$ is :

(1) $\frac{2}{3}(2\pi-3\sqrt{3})$ (2) $\frac{4}{3}(2\pi-3\sqrt{3})$

(3) $\frac{4}{3}(2\pi-\sqrt{3})$ (4) $\frac{2}{3}(4\pi-3\sqrt{3})$

13. The least value of $(\cos^2\theta-6\sin\theta\cos\theta+3\sin^2\theta+2)$ is

(1) –1 (2) $4+\sqrt{10}$

(3) $4-\sqrt{10}$ (4) 1

14. Let $I(x)=\int\frac{3dx}{(4x+6)(\sqrt{4x^2+8x+3})}$ and $I(0)=\frac{\sqrt{3}}{4}+20$. If $I\left(\frac{1}{2}\right)=\frac{a\sqrt{2}}{b}+c$, where a, b, c $\in$ N, gcd(a,b) = 1, then a + b + c is equal to :

(1) 29 (2) 28

(3) 31 (4) 30

15. The number of ways, in which 16 oranges can be distributed to four children such that each child gets at least one orange, is

(1) 429 (2) 384

(3) 403 (4) 455

16. Let A = {0, 1, 2,....9). Let R be a relation on A defined by (x, y) $\in$ R if and only if |x–y| is a multiple of 3.

Given below are two statements:

Statement I: n(R) = 36

Statement II: R is an equivalence relation.

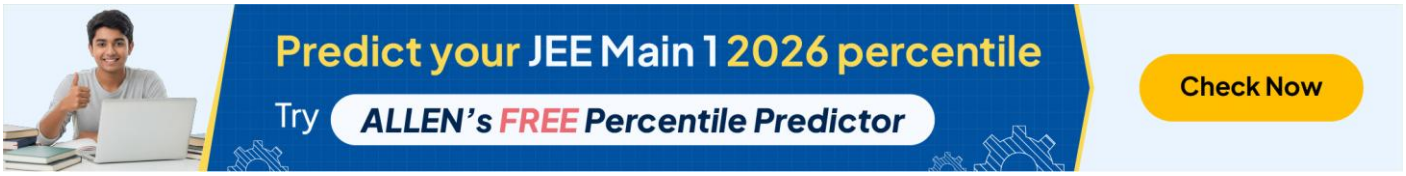
In the light of the above statements, choose the correct answer from the options given below

(1) Both Statement I and Statement II are correct

(2) Statement I is incorrect but Statement II is correct

(3) Statement I is correct but Statement II is incorrect

(4) Both Statement I and Statement II are incorrect



17. If $z=\frac{\sqrt{3}}{2}+\frac{i}{2}, i=\sqrt{-1}$, then $(z^{201}-i)^8$ is equal to

(1) –1 (2) 0

(3) 1 (4) 256

18. If the mean and the variance of the data

Class	4–8	8–12	12–16	16–20
Frequency	3	λ	4	7

are μ and 19 respectively, then the value of λ + μ is

(1) 18 (2) 21

(3) 20 (4) 19

19. An equilateral triangle OAB is inscribed in the parabola $y^2 = 4x$ with the vertex O at the vertex of the parabola. Then the minimum distance of the circle having AB as a diameter from the origin is

(1) $4(3-\sqrt{3})$ (2) $2(8-3\sqrt{3})$

(3) $4(6+\sqrt{3})$ (4) $2(3+\sqrt{3})$

20. Let $\sum_{k=1}^{n} a_k = \alpha n^2 + \beta n$. If $a_{10} = 59$ and $a_6 = 7a_1$ then $\alpha + \beta$ is equal to

(1) 12 (2) 3

(3) 5 (4) 7

SECTION-B

21. If the solution curve y = f(x) of the differential equation $(x^2 - 4)\, y' - 2xy + 2x(4 - x^2)^2 = 0$ $x > 2$, passes through the point (3, 15), then the local maximum value of f is ……

22. If the image of the point P(a, 2, a) in the line $\frac{x}{2}=\frac{y+a}{1}=\frac{z}{1}$ is Q and the of image of Q in the line $\frac{x-2b}{2}=\frac{y-a}{1}=\frac{z+2b}{-5}$ is P, then a + b is equal to……..

23. The number of elements in the

set $S=\left\{x : x \in [0,100] \text{ and } \int_0^x t^2 \sin(x-t)dt = x^2\right\}$ is.....

24. Let $A=\begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix}$ and B be a matrix such that $B(I - A) = I + A$. Then the sum of the diagonal elements of B^TB is equal to_____.

25. Let S denote the set of 4-digit numbers abcd such that a > b > c > d and P denote the set of 5-digit numbers having product of its digits equal to 20. Then n(S) + n(P) is equal to……



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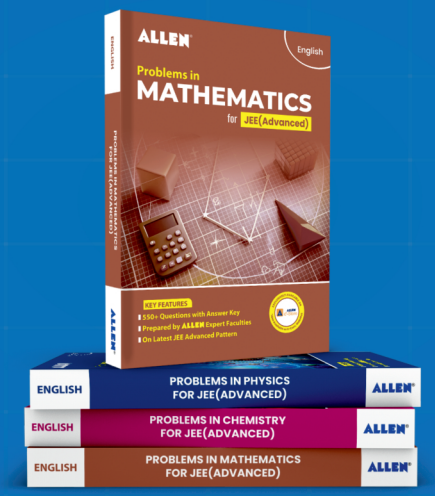


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